

Chacon Universality Gives a Full Negative Answer to Austin’s Weak-Mixing Question

Literature-implied answer packet
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Status: literature-implied answer; full negative resolution. The classical Chacon transformation is weakly mixing but cannot be the first transformation in any divergent bounded double-average example. The decisive input is Lesigne–Rittaud–de La Rue’s published theorem that Chacon is universal for weak disjointness. The short same-space diagonal deduction below does not appear to have been stated explicitly in the literature.

1 Source question

For probability-preserving transformations S, T on the same probability space and bounded measurable functions f, g , consider

$$B_N(x) := \frac{1}{N} \sum_{n=0}^{N-1} f(S^n x) g(T^n x). \quad (1)$$

Austin constructs pairs for which even the integrals of these averages fail to converge. His conclusion asks whether every weakly mixing transformation can appear, up to isomorphism, as S in a nonconvergent example [1].

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The known examples of non-convergence of the averages [\[1\]](#) provide rather diverse measure-preserving transformations in the role of S (or T). Many further questions about such transformations could be pursued. For example, it would be interesting to know whether *any* weakly mixing transformation can appear as S ; whether the ranks of S and T could help to control the averages in [\[1\]](#) (Gaussian examples always have infinite rank, and so do those in [\[14\]](#)); or which two-step distal transformations can appear.

In [\[8\]](#), Frantzikinakis and Host also pose various questions about recurrence involving non-commuting transformations. I do not think the methods of the present note come close to answering these.

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The natural universal reading has a negative answer. In fact, one classical weakly mixing system forces convergence for every possible second system and every pair of bounded observables.

2 The supporting theorem

Lesigne, Rittaud, and de La Rue define two systems (X, μ, C) and (Y, ν, T) to be *weakly disjoint* if, for every $f \in L^2(\mu)$ and $g \in L^2(\nu)$, there are full-measure sets $A \subseteq X$ and $D \subseteq Y$ such that

$$\frac{1}{N} \sum_{n=0}^{N-1} f(C^n x) g(T^n y) \quad (2)$$

converges for every $(x, y) \in A \times D$. A system is called *universal* if it is weakly disjoint from every dynamical system [2, Definition 1].

1. INTRODUCTION

1.1. Definition. In this article we call a *dynamical system* any probability measure preserving dynamical system on a Lebesgue space : a dynamical system is a quadruple $(X, \mathcal{A}, \mu, T)$ where $(X, \mathcal{A}, \mu)$ is a Lebesgue probability space and T is a measurable transformation of $(X, \mathcal{A})$ which preserves the measure μ . When there will be no ambiguity, this dynamical system will be denoted by the symbol T alone.

Definition 1. Two dynamical systems $(X, \mathcal{A}, \mu, T)$ and $(Y, \mathcal{B}, \nu, S)$ are *weakly disjoint* if, given any function $f \in L^2(\mu)$ and any function $g \in L^2(\nu)$, there exist a set A in $\mathcal{A}$ and a set B in $\mathcal{B}$ such that

$$(1.1) \quad \begin{aligned} &\bullet \mu(A) = \nu(B) = 1 \\ &\bullet \text{for all } x \in A \text{ and for all } y \in B, \text{ the sequence} \\ &\quad \left(\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \cdot g(S^n y) \right)_{N>0} \end{aligned}$$

converges.

Note that, by Birkhoff's ergodic theorem applied to the cartesian product of the dynamical systems $T \times S$, we know that for $\mu \otimes \nu$ -almost all (x, y) the sequence (1.1) converges. But a set of full measure for the product measure does not necessary contain a "rectangle" $A \times B$ of full measure.

1.2. Motivation. The weak disjointness concept appears for the first time in [21] under the name "propriété ergodique produit forte". The aim was to study a well known open problem in pointwise ergodic theory: given two commuting measure preserving transformations T and S of the same probability space $(X, \mathcal{A}, \mu)$, is it true that for any functions f and g in $L^2(\mu)$, the sequence

$$\left(\frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) \cdot g(S^n x) \right)_{N>0}$$

converges for μ -almost all x ? (If T and S are weakly disjoint, the answer is positive.)

This weak disjointness property is also interesting to study for the following reasons.

- It defines a new invariant in the theory of metric isomorphisms of dynamical systems. (If T and S are weakly disjoint and if T' is a measure theoretic factor of T , then T' and S are weakly disjoint.)
- It has strong links with the rich theory of joinings in Ergodic Theory.
- It gives an opportunity to describe an interesting variety of examples.

1.3. Brief description of the content. Let us give a few useful definitions.

Definitions. If a dynamical system is weakly disjoint from itself, we say that this dynamical system is *self-weakly disjoint*. A natural generalization of the notion of weak disjointness of a pair of dynamical systems is the notion of weak disjointness of a finite family of dynamical systems (see the discussion at the end of Section 2). If k copies of a given dynamical systems are weakly disjoint, we say that this dynamical system is *self-weakly disjoint of order k* . Finally a dynamical system is called *universal* if it is weakly disjoint from any dynamical system.

The full rectangle in this definition is substantially stronger than ordinary $\mu \otimes \nu$ -almost-everywhere convergence: a product-null diagonal can be inserted into the rectangle after the two systems are placed on the same space.

In Section 4.2 the authors first recall that Chacon's transformation is weakly mixing (but not mixing), and then prove the following statement.

4.2. Chacon's transformation. The Chacon dynamical system is a well known example of a weakly mixing but not mixing system. We denote by $(X, \mathcal{A}, \mu, T)$ an invertible regular version of this system. From the result proved in [4] we deduce directly the existence of a subset X_0 of full measure in X such that every point in X_0 is generic, and such that for all x and y in X_0 , either there exist $p \in \mathbb{Z}$ such that $T^p x = y$, or $\Delta_N(x, y) \rightarrow \mu \otimes \mu$, when $N \rightarrow \infty$. Of course, if x is generic and if $T^p x = y$, then $\Delta_N(x, y)$ goes to the image under $Id \times T^p$ of the diagonal measure on X^2 .

Therefore this dynamical system is self-weakly disjoint. In fact, we have more.

Proposition 4.2. *The Chacon dynamical system is universal.*

Proof. By Theorem 3.7, it is sufficient to prove that the Chacon dynamical system is self-weakly disjoint of all orders.

Let k be an integer ≥ 2 , and $x_1, x_2, \dots, x_k$ in X_0 . Let λ and λ' be two limit values of the sequence $(\Delta_N(x_1, x_2, \dots, x_k))$.

Since every point in X_0 is generic, λ and λ' are two joinings of k copies of T . Since for all $(x, y) \in (X_0)^2$ the sequence $(\Delta_N(x, y))$ converges, the restrictions of λ and λ' to any sub- σ -algebra generated by two coordinates always coincide. (These restrictions are either the product measure $\mu \otimes \mu$, or of the form Λ_p , where $\Lambda_p(A \times B) := \mu(A \cap T^{-p}B)$.) But we know that Chacon's dynamical system has minimal self-joinings of all orders ([5]), and we conclude that $\lambda = \lambda'$. This proves that the system is self-weakly disjoint of all orders. $\square$

In fact, the proof of Theorem 3.7 shows that, in order to prove that an ergodic

Thus Proposition 4.2 of [2] says exactly that Chacon's transformation is weakly disjoint from every probability-preserving dynamical system.

3 The diagonal consequence

Theorem 1. *Let $(X, \mathcal{B}, \mu, C)$ be an invertible probability-preserving Chacon system. For every probability-preserving transformation T of the same Lebesgue probability space and every $f, g \in L^\infty(\mu)$, the averages*

$$B_N(x) = \frac{1}{N} \sum_{n=0}^{N-1} f(C^n x) g(T^n x)$$

converge almost everywhere and in $L^p(\mu)$ for every finite $p \geq 1$. Consequently the scalar averages

$$I_N = \frac{1}{N} \sum_{n=0}^{N-1} \int_X f(C^n x) g(T^n x) d\mu(x)$$

converge. The same conclusions hold when the first system is any system measurably isomorphic to Chacon's transformation.

Proof. Fix T, f, g . By Proposition 4.2 and Definition 1 of [2], there are sets $A, D \in \mathcal{B}$ with $\mu(A) = \mu(D) = 1$ such that the two-point averages (2) converge for every $(x, y) \in A \times D$. Here

the first and second systems act on the same underlying space, so $E := A \cap D$ has full measure. For every $x \in E$, substitute $y = x$ in (2). This gives exactly the almost-everywhere convergence of B_N .

Let $M = \|f\|_\infty \|g\|_\infty$. Then $|B_N| \leq M$ for all N , and the almost-everywhere limit B also satisfies $|B| \leq M$. Dominated convergence applied to $|B_N - B|^p$ gives $B_N \rightarrow B$ in every finite L^p . Finally,

$$I_N = \int_X B_N d\mu \longrightarrow \int_X B d\mu,$$

which proves convergence of the integrated averages.

Weak disjointness is invariant under metric isomorphism [2, Introduction]. Equivalently, transporting T, f, g through an isomorphism reduces an arbitrary copy of Chacon’s system to the case just proved. $\square$

4 Resolution, provenance, and checks

Chacon’s transformation is weakly mixing, while Theorem 1 shows that it cannot appear as S in a nonconvergent example of (1). This answers Austin’s universal question negatively. It also removes any ambiguity about the convergence mode: pointwise almost-everywhere convergence, every finite- L^p convergence, and convergence of the scalar integrals all hold.

The supporting theorem predates Austin’s question by more than two decades, so its authors do not claim to answer that question. The relation is a direct agent-identified implication and is therefore recorded as a *literature-implied answer*, not as a new counterexample or an explicit later literature answer.

A bounded novelty check through 9 August 2026 searched the run indexes; exact title, arXiv-id, and question-wording matches; citation and title searches combining Austin’s paper with Chacon, universality, and weak disjointness; and recent related counterexample papers. No later source explicitly stating this corollary was found. Novelty confidence is intentionally low because the only substantive dynamical input is the published Proposition 4.2.

Human-review recommendation. Verify the natural universal reading of Austin’s word “any” and the full-rectangle quantifiers in Definition 1. Once those are fixed, the diagonal substitution and dominated-convergence steps are immediate.

References

- [1] T. Austin, *Non-convergence of some non-commuting double ergodic averages*, Proc. Amer. Math. Soc. **153** (2025), 1701–1707; arXiv:2407.08630.
- [2] E. Lesigne, B. Rittaud, and T. de La Rue, *Weak disjointness of measure preserving dynamical systems*, Ergodic Theory Dynam. Systems **23** (2003), 1173–1198, doi:10.1017/S0143385702001505; HAL:hal-00017086.